Worksheet: Simple and Compound Interest

Group names: _____

1. The purpose of this problem is to highlight the importance of reading carefully. Subtle differences in wording can mean very different things!

   For this problem, assume an investment has an initial deposit (principal) of $1600 and no other deposits or withdrawals are made.

   Your work should include the expression or calculation you put into the spreadsheet/calculator as well as the numerical answer.

   (a) 5% of the initial deposit is...

   calculation: _____ answer: _____

   (b) The account increased by 5% means the account gained...

   calculation: _____ answer: _____

   (c) The account increased by 5% means the new balance in the account is...

   calculation: _____ answer: _____

   (d) The new balance is 105% of the original balance means the new balance is...

   calculation: _____ answer: _____

   (e) Your final numerical answers in parts (c) and (d) should be the same though your work/calculations should be different. Answer the following by doing **both** calculations.

   The account increased by 20% means the new balance is...

   calculation: _____ answer: _____

   calculation: _____ answer: _____

   The account increased by 100% means the new balance is...

   calculation: _____ answer: _____

   calculation: _____ answer: _____

   (f) What is a more common way of saying an account "increased by 100%"?

   (g) Suppose you are told that each month the new balance is 104% of the old balance. What percent of the old balance is **gained** each month?

Recall:

P = principal / starting amount	r = annual interest rate (APR)
I = interest	n = number of compounding periods per year
A = final amount / future value	t = number of years

Future Value $A = P + I$

Simple Interest over Time $A = P(1 + rt)$

Compound Interest over Time $A = P(1 + \frac{r}{n})^{nt}$

For the following problems,

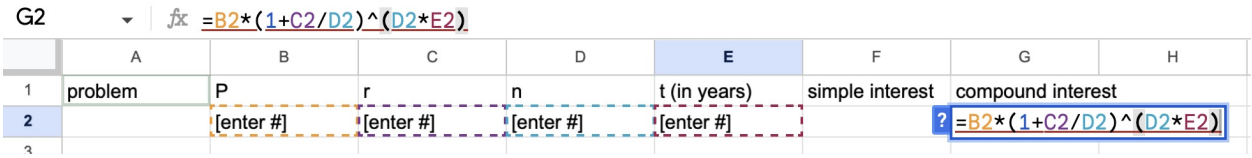
(i) Identify the value for each variable and which formula you should use

(ii) use a **spreadsheet** or a calculator to compute the quantities.

For each quantity, carefully write **what you need to compute** along with the answer.

To be efficient, write the following spreadsheet formula for compound interest (you would put in the correct numbers for each problem where it says [enter number]). You can fill down to compute the correct total amount for each problem. Write a similar formula for simple interest.

Copy and paste the rows for each problem.



Example: Your uncle gives you a simple interest loan of \$500 for one year, at 4% annual interest.

(a) How much interest do you owe?

(b) What is the total amount you will owe him at the end of the year?

Answer:
This is a simple interest problem. $P = 500$, $r = 0.04$, $t = 1$.
(a) Total interest: $I = Prt = 500 * 0.04 * 1 = \$20$
(b) Final amount: $A = 500(1 + 0.04) = \$520$

2. You borrowed \$1500 from a relative.

(a) Suppose she charged you 5% APR, **compounded** monthly. If you paid her back 5 years later, how much money did you give her?

$P =$ _____ $r =$ _____ $n =$ _____ $t =$ _____

$A =$ _____

(b) In the above situation, how much more interest did you pay than if she had charged you simple interest for 5 years?

3. You got a bonus of \$7,500 and you want to start a college fund for your child. You find an account paying 9.75% APR compounded quarterly. If your child just turned two years old, how much will you have when they turn 18? How much of that account balance is interest?

   $P =$ _____ $r =$ _____ $n =$ _____ $t =$ _____

   $A =$ _____ total interest = _____

4. Calculate how much you would have in the previous problem if it was compounded daily instead of quarterly. How much additional interest did you earn from the change in the number of compounding periods?

   $P =$ _____ $r =$ _____ $n =$ _____ $t =$ _____

   $A =$ _____ total interest = _____

   additional interest = _____

   To determine the **effective annual interest rate**, compute the total interest I paid for **one year**, compute I/P, and convert it into a percentage, either by multiplying by 100 or clicking on the **%** in the toolbar. (Recall since $A = P + I$, $I = A - P$.)

5. If you are considering a credit card with an APR of 27.49%, compounded **daily**, what is the **effective annual interest rate**?

6. Which is better: an account that earns 7.25% compounded quarterly, or an account that earns 7.15% compounded daily? Give the **effective annual interest rate** for each account, and explain what that tells you about which is better.

7. How much would you need to deposit today to end up with one million dollars if you can find an account that pays 10% interest, compounded daily, for 50 years?

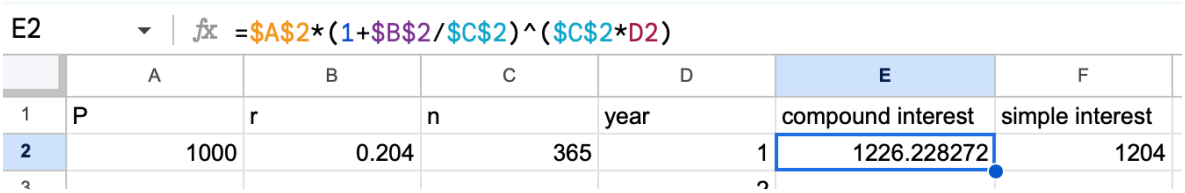
   Note that if $A = P(1 + \frac{r}{n})^{nt}$ then by algebra, $P = \dfrac{A}{(1 + \frac{r}{n})^{nt}}$.

Optional: Compare Simple and Compound Interest with Charts

8. For this problem, we want to compare simple interest to compound interest over time, in a graph.

   Scenario: You put $1000 on a credit card, which charges 20.4% APR, compounded daily. Suppose this is an unrealistic credit card that does not have monthly minimum payments (we will talk about how to handle those later).

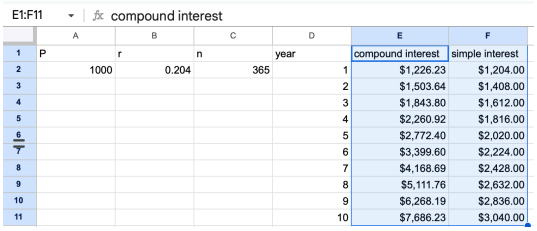
   (a) Make a new spreadsheet with spaces for P, r, n, and t (which we will label "years"). Make columns for compound interest and simple interest. I am showing you the formula for compound interest in the screenshot; you will need to modify it to enter simple interest.



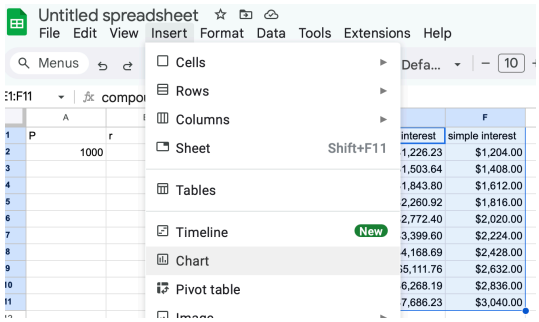
	A	B	C	D	E	F
1	P	r	n	year	compound interest	simple interest
2	1000	0.204	365	1	1226.228272	1204

   (b) Fill down the "year" column by typing in 1 and 2, selecting both, and filling down until you reach 10 years. Then fill down the compound interest and simple interest columns.

   (c) Select the entries in the Compound Interest and Simple Interest columns, go to the `Insert` column, select `Chart`.



	A	B	C	D	E	F
1	P	r	n	year	compound interest	simple interest
2	1000	0.204	365	1	$1,226.23	$1,204.00
3				2	$1,503.64	$1,408.00
4				3	$1,843.80	$1,612.00
5				4	$2,260.92	$1,816.00
6				5	$2,772.40	$2,020.00
7				6	$3,399.60	$2,224.00
8				7	$4,168.69	$2,428.00
9				8	$5,111.76	$2,632.00
10				9	$6,268.19	$2,836.00
11				10	$7,686.23	$3,040.00



   (d) Move the chart so that you can see the chart and the data, and experiment with what happens to the chart if you change the number of compounding periods, or the interest rate. What do you notice?

   (e) How much do you owe in interest at the end of 10 years?

   Simple interest: _____ Compound interest: _____

9. Modify the previous question to make a chart that shows the difference in interest if you compound $1000 daily versus annually at 6% annual interest rate for 100 years, looking at values every 10 years. You will need to have two different values for n and two different compound interest columns.

   What do you notice?